

Tips on creating a professional log file for problem sets

This handout provides some tips on how to create a professional-looking log file for your problem sets. Think of the log file as a readable record of your analysis, not a data dump.

1. **Do-file** Create a do-file that will include all problem set questions, Stata syntax, results, and written interpretation.
2. **Log file** Logs can be saved as text (.txt) files or in Stata markup language (.smcl) format. I prefer txt for its simplicity and portability.

```
capture log close           // ensures no log is already open
log using C:\data\yourlogname.txt, text // opens log file and begins logging
...
log close                   // stops logging and saves log file
```

The command `capture` before another Stata command tells Stata to ignore any error message that might result. For example, if there were no log file open, `log close` would produce an error message.

3. **Header** Include a clear and descriptive header at the top. For this to appear in your log file, it must come after the `log using` command.

```
/******
* LPO 8800: Problem Set 1
* Name: Sean P. Corcoran
* Date: August 25, 2026
* Topic: Descriptive statistics
*****/
```

4. **Comments** Your do-file can include comments that Stata ignores. Multi-line comments are enclosed in `/* */` (like the header above). Single-line comments begin with `*` or `//`. The main body of your do-file should include the problem set questions (copied and pasted from the original, in comments), Stata syntax that relates to each question, and written interpretation (commented).
5. **Section headers** Use section headers that correspond to the problem set questions. This really helps with readability.

```
/******
* QUESTION 1: MEANS AND STANDARD DEVIATIONS
*****/
```

6. **Avoid text wrapping** Stata will wrap lines whenever text extends beyond the page width. The result can be unpredictable and look unprofessional. At least two factors affect this: (1) the width of the Results window on your screen, and (2) the maximum linesize. Text may wrap because your Results

window is too small; then, what you see in your window is what you get in the log. Widen this window to accommodate the maximum linesize. Use `set linesize` to choose a maximum width for a line of text. I typically use 80 characters for my page width, but you can adjust up or down as needed. The do-file editor includes a visual reference line at 80 characters; this can also be adjusted in Preferences.

```
set linesize 80
```

Relatedly, the text for your problem set questions and interpretation may extend beyond the maximum line width. Break these up into multiple lines to avoid text wrapping:

```
* The data are a sample of classroom teachers from a large urban school district
* in 2006. The variables include characteristics of the teacher (e.g., total
* years of experience, master's degree attainment), average student test scores
* in the class (reading, math, language, science, and social studies), and
* characteristics of students in the classroom (e.g., proportion in special
* education, proportion immigrant, proportion by race/ethnicity). Note the...
```

The same applies for Stata commands which can sometimes be much longer than the maximum line width. Placing three forward slashes at the end of a line tells Stata that the command will continue on the next line. This allows you to break up a command and avoid text wrapping. For example:

```
twoway (connected pdc year, mlabel(pdc) mlabsize(vsmall) mlabposition(12)) ///
      (connected pdc_snap year if year>=2015, ///
      mlabel(pdc_snap) mlabsize(vsmall) mlabposition(6)), ///
      ylabel(0(5)60, format(%2.0f) ticks tcolor(black) tposition(outside)) ///
      xlabel(2012(1)2025, ticks tcolor(black) tposition(outside)) ///
      legend(rows(1) position(6) order(1 "DC" 2 "DC via SNAP") region(lwidth(none))) ///
      xtitle(, size(zero)) graphregion(fcolor(white)) xsize(7) ysize(5) name(dc, replace)
```

- 7. Suppressing results** Sometimes you have comments, syntax, or results that you do not want to appear in your resulting log file. There are several ways to accomplish this: (1) you can temporarily turn the log off (`log off`) and back on again (`log on`); or (2) you can put the prefix `quietly:` before a command. The latter will suppress any results from the Results window and the log file.

Again, the log file should be a readable record of your analysis. Keep exploratory clutter out of the final log (e.g., experiments with syntax, describe and browse commands, etc.). What is contained in the log should be your final product, a *successful, reproducible run of the finished analysis*.

- 8. Exporting to PDF** Following the (`log close`) command, your log file will be saved as a `.txt` or `.smcl` file. You can then convert this within Stata to a PDF file using `translate`:

```
translate yourlogname.txt yourlogname.pdf, translator(txt2pdf) ///
      cmdnumber(off) logo(off) header(off) replace
```

Note the options that suppress the line numbers, Stata logo, and headers. I find this produces a cleaner, more professional log file. If you find that your `txt` file is overflowing the page width (e.g., unexpected wrapping), you can adjust the margins of the PDF in the options:

```
translate yourlogname.txt yourlogname.pdf, translator(txt2pdf) ///
      cmdnumber(off) logo(off) header(off) lmargin(0.5) rmargin(0.5) replace
```

If you're having trouble getting the txt file to translate to PDF properly or have other Mac-vs-PC issues, see the `help translate` documentation. You can also use other methods (outside of Stata) to print the txt file to a PDF.

9. **Graphs** Stata does not embed graphic images in log files. One approach is to save graphs as PDFs (or other image files) that can be later combined with your log PDF:

```

histogram achmat08
graph export histogram.pdf, as(pdf) replace

```

If you install a small app called PDF Toolkit the following command will combine PDFs directly in Stata. The command invokes the command line in Windows and combines the PDFs. (It may or may not work for Mac). See also the `putpdf` solution below.

```
!pdftk PS1_Corcoran.pdf histogram.pdf cat output PS1_Corcoran_final.pdf
```

10. **Other options** Stata has many tools for exporting and formatting results that go beyond these basics. See `putdocx`, `putexcel`, and `putpdf` for a suite of commands that allow you to build a Word, Excel, or PDF document that includes text, tables, and figures. These are powerful, but a bit limiting in that your do-file has to specify each element to include in the document. The user-written command `graphlog` is a nice alternative that uses a $\text{\LaTeX}$ installation to combine Stata logs and graphics (Hansen, 2015) Another user-written command `markdoc` (Haghighi, 2016) does something similar.

The following is a (somewhat awkward) solution using `putpdf` that combines a log file and one graph into a a single PDF:

```

log using logfile.txt, text    // opens log file and begins logging
...                          // [everything here that you want in the log file]
log close                    // stops logging and closes log file

putpdf begin                  // starts creating a PDF

putpdf paragraph, font("Courier", 8) // starts a paragraph using Courier size 8 font
file open mylog using logfile.txt, read text // opens log file file
file read mylog line
while r(eof)==0 {            // reads log line by line and puts in PDF
    putpdf text ("`line'"), linebreak
    file read mylog line
}
file close mylog

putpdf pagebreak              // page break
putpdf paragraph, halign(center) // starts new (centered) paragraph
putpdf image "histogram.png", width(5) // inserts the graphic

putpdf save "myresults.pdf", replace // saves the PDF

```